

Phase 4: Segment Profiling

& SHAP Explainability — Detailed Technical Report

1. What Is Segment Profiling?

After K-Means assigned each of the 93,357 customers to one of 3 clusters (Phase 3), the clusters are just numbered labels — Cluster 0, 1, 2 — with no business meaning. Phase 4 transforms these numerical labels into actionable customer segments by:

- **Building statistical profiles:** Computing the mean, median, and standard deviation of raw RFM values (Recency, Frequency, Monetary) per cluster — translating abstract cluster assignments back into interpretable business metrics.
- **Assigning human-readable labels:** Using normalised RFM thresholds to assign names like 'Champions', 'Potential Loyalists', and 'Hibernating' — making each cluster immediately understandable to a non-technical stakeholder.
- **Explaining with SHAP:** Using SHAP (SHapley Additive exPlanations) to quantify which RFM feature drives each cluster's identity — providing model-agnostic feature importance at the per-segment level.

1.1 Why Is Profiling Necessary?

Clustering algorithms optimise mathematical objectives (minimise WCSS, maximise silhouette) but do not assign meaning. A marketing team cannot act on 'Cluster 2' — they need to know that Cluster 2 represents 'customers who purchased once over a year ago and may have churned'. Profiling bridges the gap between unsupervised learning output and business-actionable intelligence.

1.2 Pipeline Context

Phase 4 receives two inputs:

- **clustered_customers.csv:** 93,357 rows with customer_id, PC1, PC2, and cluster label (0, 1, or 2) — the output of K-Means from Phase 3.
- **rfm_raw.csv:** 93,357 rows with customer_id, recency (days since last purchase), frequency (order count), and monetary (total spend in BRL) — the raw RFM features from Phase 1, before any transformation or scaling.

By merging these on customer_id, we can compute raw RFM statistics per cluster and assign labels based on the actual business meaning of each cluster's customers.

2. Profiling Methodology

2.1 Building the Profile Table

The profiling function merges clustered customers with raw RFM values and computes aggregate statistics per cluster:

```
merged = clustered.merge(rfm_raw, on='customer_id')
profile = merged.groupby('cluster').agg(
    recency = ('recency', 'mean'),
    frequency = ('frequency', 'mean'),
    monetary = ('monetary', 'mean'),
    count = ('customer_id', 'count')
)
```

This gives us the mean RFM values per cluster, which we use to understand each segment's behavioural profile:

Cluster	Recency (days)	Frequency (orders)	Monetary (BRL)	Count	% of Total
0	155.0	1.0	153.7	42,865	45.9%
1	268.6	2.0	294.7	2,801	3.0%
2	405.0	1.0	156.0	47,691	51.1%

2.2 Interpreting the Raw Profiles

- **Cluster 0 (42,865 customers):** Low recency (155 days = recent), single purchase (frequency 1.0), moderate spend (BRL 154). These are customers who bought recently but only once — potential repeat buyers.
- **Cluster 1 (2,801 customers):** Medium recency (269 days), the only cluster with repeat purchases (frequency 2.0), and the highest spend (BRL 295). These are the most valuable customers.
- **Cluster 2 (47,691 customers):** High recency (405 days = long time ago), single purchase (frequency 1.0), moderate spend (BRL 156). These are lapsed customers who may have churned.

2.3 Segment Heatmap

The heatmap below visualises the normalised RFM profiles with raw value annotations. Colour intensity represents the min-max normalised value (0 to 1) across clusters for each feature, while the text annotations show the actual raw means.

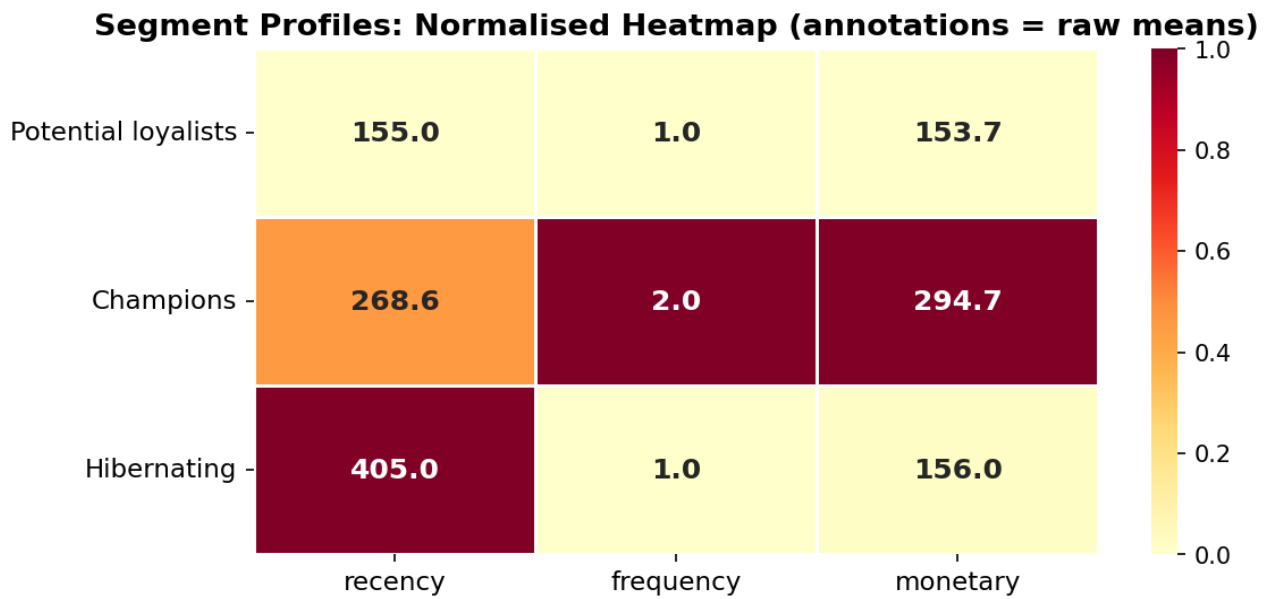


Figure 1: Normalised segment heatmap. Champions stand out with the highest frequency and monetary values (dark cells). Hibernating has the highest recency (darkest cell in recency column).

Note: In the heatmap, 'higher' always means 'more extreme'. For recency, a higher value means more days since last purchase (i.e., less recent). For frequency and monetary, higher means more purchases and more spending — which is desirable.

3. Segment Labelling — Normalised Threshold Approach

Raw cluster numbers (0, 1, 2) are meaningless to business stakeholders. We assign human-readable labels using a normalised threshold system that evaluates each cluster on two dimensions: **recency** (how recently they purchased) and **value** (how much they purchase).

3.1 How Normalisation Works

For each cluster, we compute two normalised scores between 0 and 1:

Recency Score (r_norm) — Inverted

$$r_norm = 1 - (\text{mean_recency} - \text{min_recency}) / (\text{max_recency} - \text{min_recency})$$

Recency is **inverted** because lower recency (fewer days since last purchase) is better. After inversion, $r_norm = 1$ means the most recent cluster, $r_norm = 0$ means the least recent.

Value Score (v_norm) — Direct

$$\begin{aligned} f_norm &= (\text{mean_frequency} - \text{min_freq}) / (\text{max_freq} - \text{min_freq}) \\ m_norm &= (\text{mean_monetary} - \text{min_mon}) / (\text{max_mon} - \text{min_mon}) \\ v_norm &= (f_norm + m_norm) / 2 \end{aligned}$$

Value is the average of normalised frequency and normalised monetary. Higher = more valuable. This captures both purchase volume and spend amount in a single metric.

3.2 Computed Scores

Cluster	Mean Recency	r_norm (inverted)	Mean Frequency	f_norm	Mean Monetary	m_norm	v_norm (avg F+M)
0	155.0	1.00	1.0	0.00	153.7	0.00	0.00
1	268.6	0.55	2.0	1.00	294.7	1.00	1.00
2	405.0	0.00	1.0	0.00	156.0	0.02	0.01

3.3 Label Assignment Rules

A threshold of **0.5** is applied to both scores:

Condition	Label	Meaning
$r_norm \geq 0.5$ AND $v_norm \geq 0.5$	Champions	Recent AND high value — best customers
$r_norm \geq 0.5$ AND $v_norm < 0.5$	Potential Loyalists	Recent but low value — nurture opportunities
$r_norm < 0.5$	Hibernating	Not recent — lapsed customers

3.4 Applying Rules to Our Clusters

- **Cluster 0:** $r_norm = 1.00$ (≥ 0.5 ✓), $v_norm = 0.00$ (< 0.5 ✗) → **Potential Loyalists**. These are the most recent buyers, but they've only purchased once with moderate spend. They have the potential to become Champions with the right re-engagement.
- **Cluster 1:** $r_norm = 0.55$ (≥ 0.5 ✓), $v_norm = 1.00$ (≥ 0.5 ✓) → **Champions**. Both recency and value are above threshold. These are the repeat buyers with the highest monetary value — the most valuable 3%.
- **Cluster 2:** $r_norm = 0.00$ (< 0.5 ✗) → **Hibernating**. Recency alone disqualifies them — regardless of value, a customer who last purchased 405 days ago is lapsed. (Their value score is also near zero at 0.01.)

3.5 Labelling Decision Chart

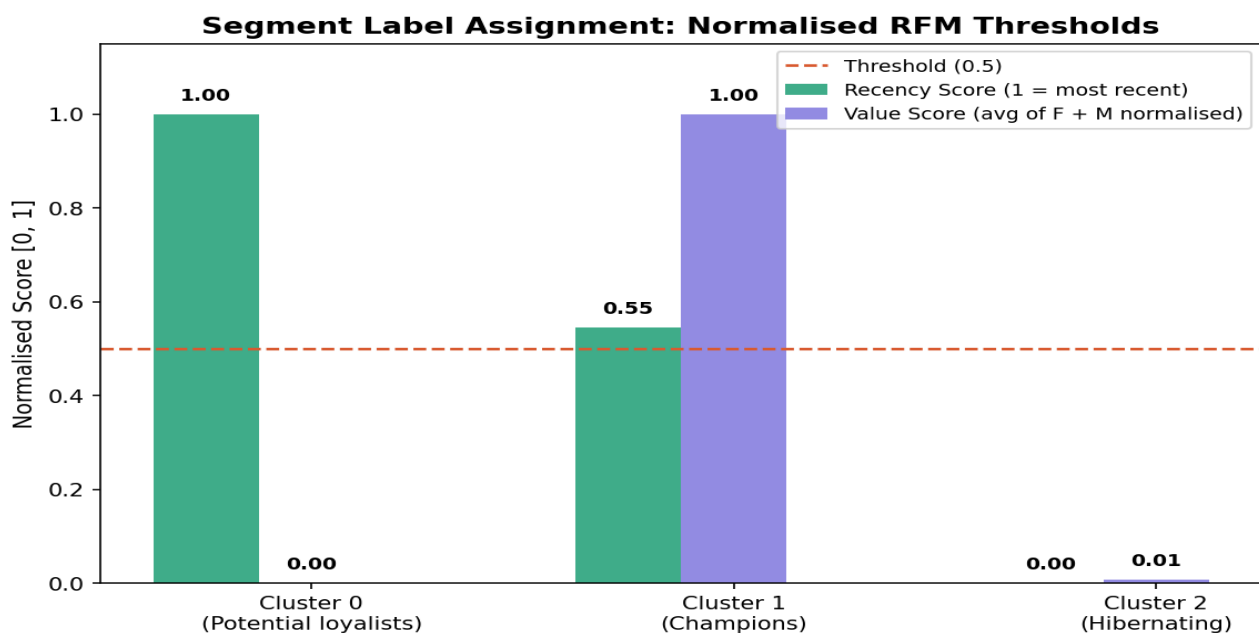


Figure 2: Normalised recency and value scores per cluster. The dashed red line at 0.5 is the threshold. Cluster 0 crosses the threshold on recency only (→ Potential Loyalists). Cluster 1 crosses both (→ Champions). Cluster 2 crosses neither (→ Hibernating).

Note: The threshold of 0.5 is a natural midpoint on the normalised [0, 1] scale. It divides clusters into 'above average' and 'below average' relative to the cluster population. This approach is robust: it does not depend on rank ordering (which can break with ties) or absolute thresholds (which vary across datasets).

4. SHAP Explainability

Cluster labels tell us *what* each segment looks like. SHAP (SHapley Additive exPlanations) tells us *why* — which RFM features are most important for distinguishing each segment from the others.

4.1 Why SHAP?

K-Means itself has no built-in feature importance measure. Unlike supervised models (e.g., Random Forest), unsupervised algorithms do not have a direct concept of 'this feature matters more'. We need an external method to explain the clusters.

SHAP is based on Shapley values from cooperative game theory — a principled way to distribute a prediction among its input features. For each prediction, SHAP assigns a value to each feature representing how much that feature contributed to pushing the prediction away from the baseline (average prediction).

Property	Description
Local accuracy	SHAP values for a prediction sum to the difference between prediction and baseline
Consistency	If a feature's contribution increases in a new model, its SHAP value won't decrease
Missingness	Missing features get SHAP value = 0
Additivity	$f(x) = \text{base_value} + \sum \text{SHAP}_i(x)$

4.2 The Proxy Model Approach

Since SHAP cannot be applied directly to K-Means, we train a **proxy classifier** that learns to predict the K-Means cluster labels from the raw RFM features. If the proxy is accurate, its SHAP values faithfully represent which features drive the clustering.

Proxy Model: Random Forest Classifier

- **Model:** RandomForestClassifier from scikit-learn
- **n_estimators = 200** (200 decision trees in the ensemble)
- **max_depth = 10** (each tree can be up to 10 levels deep)
- **random_state = 42** (reproducibility)
- **Input features:** recency, frequency, monetary (raw, unscaled values)
- **Target:** K-Means cluster label (0, 1, or 2)

Why Random Forest?

- Non-linear decision boundaries — captures complex RFM interactions
- Compatible with SHAP's TreeExplainer — exact (not approximate) SHAP values
- Robust to feature scale — raw RFM values can be used without standardisation
- Fast training — 200 trees on 93K samples takes < 2 seconds

4.3 Proxy Model Validation

The proxy is only useful if it faithfully reproduces the K-Means assignments. We validate with 5-fold stratified cross-validation:

Fold	Accuracy
1	0.9999
2	0.9999
3	0.9999
4	1.0000
5	0.9998
Mean ± Std	0.9999 ± 0.0001

Accuracy = 99.99% — the Random Forest nearly perfectly reproduces the K-Means cluster assignments from raw RFM features. This means the proxy model's SHAP values are a faithful representation of the original clustering's feature dependencies.

Note: An accuracy this high is expected. K-Means assigns clusters based on distances in PCA space, which is a linear transformation of the standardised RFM features. A 200-tree Random Forest can easily learn these decision boundaries. The near-perfect accuracy confirms that no information was lost — the proxy is a reliable stand-in for the K-Means model.

5. SHAP Feature Importance — Results

5.1 How SHAP Values Were Computed

After training the proxy Random Forest on all 93,357 customers, we use SHAP's **TreeExplainer** — an exact algorithm for tree-based models that computes SHAP values in polynomial time (not the exponential cost of brute-force Shapley computation).

```
explainer = shap.TreeExplainer(rf_model)
shap_values = explainer.shap_values(X_rfm) # shape: (93357, 3, 3)
# → 93,357 customers × 3 features × 3 classes
```

For each customer, TreeExplainer produces a 3×3 matrix: one SHAP value per feature per class. The mean absolute SHAP value per feature per class gives the global feature importance for each segment.

5.2 SHAP Importance Table

Segment	Recency SHAP	Frequency SHAP	Monetary SHAP	Dominant Feature
Potential Loyalists	0.4777	0.0279	0.0107	Recency (92.6%)
Champions	0.0008	0.0567	0.0056	Frequency (89.9%)
Hibernating	0.4784	0.0288	0.0076	Recency (92.9%)

5.3 Interpreting the SHAP Results

Recency Dominates Potential Loyalists and Hibernating

The two largest segments — Potential Loyalists (45.9%) and Hibernating (51.1%) — are distinguished almost entirely by recency (SHAP = 0.478 for both). This makes sense: both segments have frequency = 1.0 and similar monetary values (BRL 154 vs 156), so the only axis separating them is when they last purchased.

- **Potential Loyalists** have a mean recency of 155 days — they bought within the last ~5 months.
- **Hibernating** have a mean recency of 405 days — they bought over a year ago.
- Recency's SHAP value of ~0.48 means it shifts the prediction probability by 48 percentage points toward these segments.

Frequency Dominates Champions

Champions are the only repeat buyers (frequency = 2.0 vs 1.0 for the other segments). Frequency has a SHAP value of 0.0567 for Champions — apparently small in absolute terms, but it accounts for 89.9% of the total importance for this segment. Recency's SHAP for Champions is almost zero (0.0008) because Champions span a wide range of recency values (mean 269 days with high

variance).

Monetary Has Minimal Impact

Monetary value contributes < 2% of SHAP importance for every segment. This does not mean monetary doesn't matter — it means that after controlling for recency and frequency, monetary adds little *additional* discriminative power. In this dataset, monetary is highly correlated with frequency (repeat buyers spend more), so frequency already captures most of the monetary signal.

5.4 SHAP Heatmap

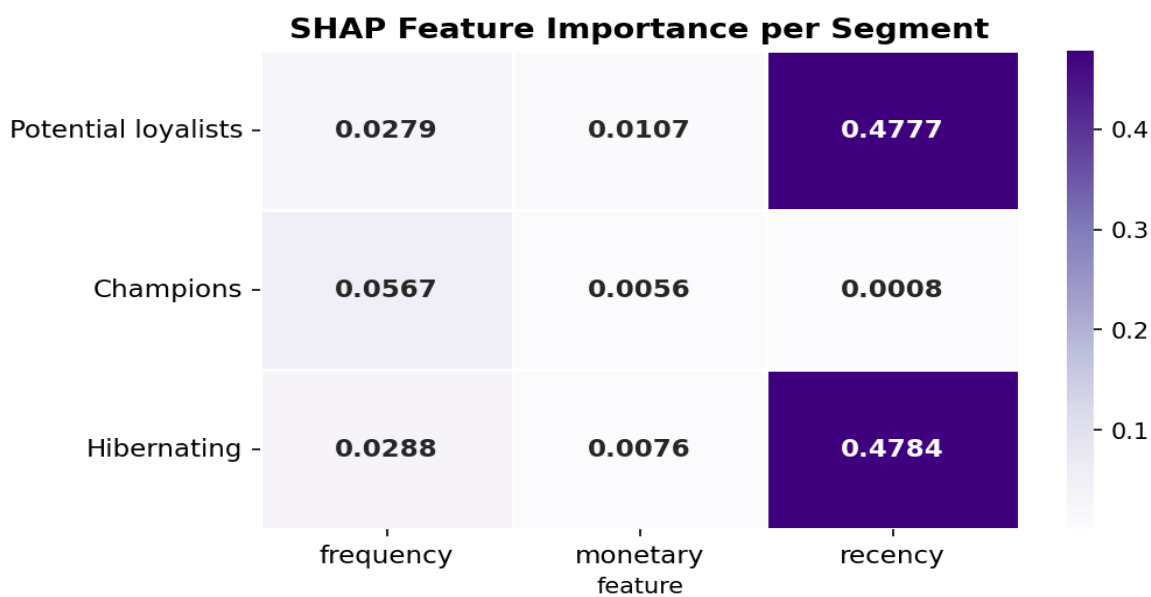


Figure 3: SHAP feature importance per segment. Recency (rightmost column) dominates Potential Loyalists and Hibernating with deep purple cells. Frequency dominates Champions (lighter purple in left column).

5.5 SHAP Bar Chart

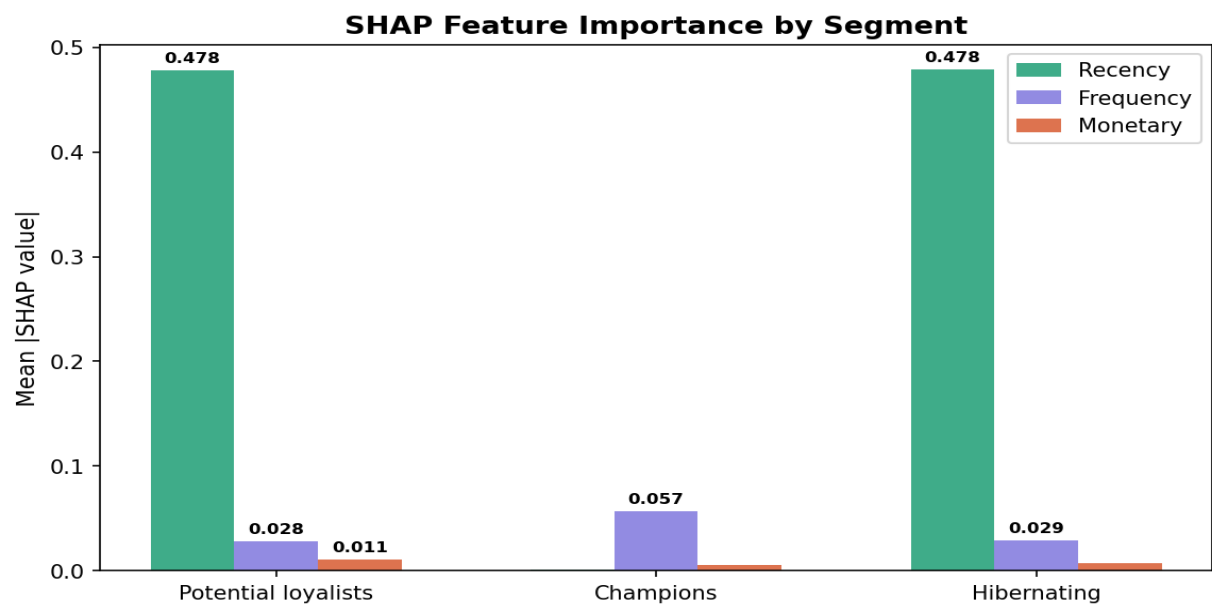


Figure 4: Grouped bar chart of SHAP importance. The recency bars for Potential Loyalists and Hibernating tower over all other values, confirming recency is the primary clustering axis.

6. RFM Distributions per Segment

The box plots below show the distribution of raw RFM values within each segment. Outliers are hidden for clarity (they were capped at the 1st/99th percentiles in Phase 1, so extreme values are bounded).

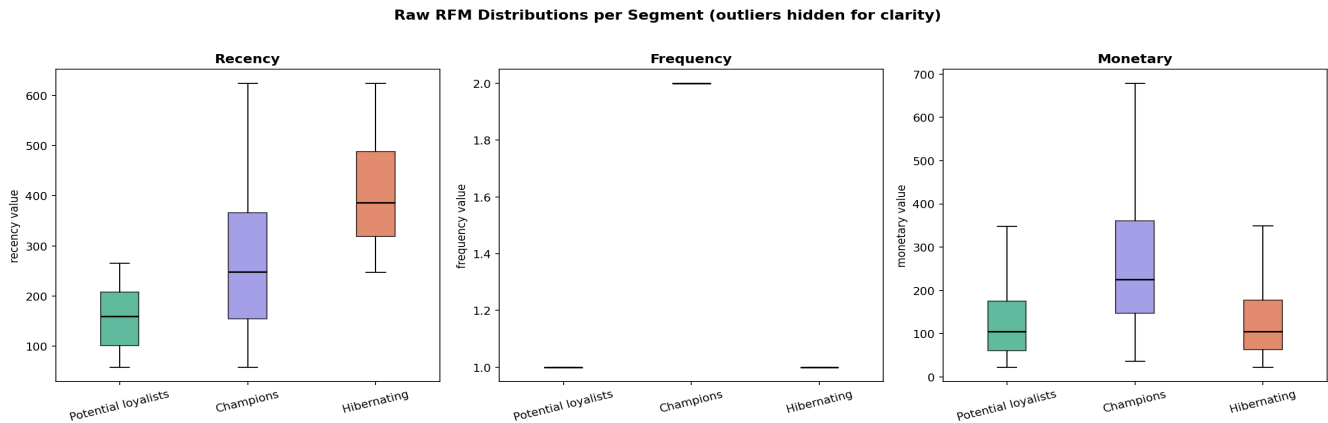


Figure 5: Box plots of raw Recency, Frequency, and Monetary per segment. Outliers suppressed. Whiskers extend to the 1st/99th percentile values.

6.1 Recency Distribution

- **Potential Loyalists:** Tight distribution between ~55 and ~280 days (median ~155). All members purchased within the last 9 months.
- **Champions:** Wide spread from ~50 to ~640 days (median ~245). Champions are defined by repeat purchases, not by recency — some Champions last bought recently, others long ago.
- **Hibernating:** Distribution starts above ~280 days and extends to ~640 days (median ~400). These are uniformly old purchasers.
- **Key insight:** The Potential Loyalists and Hibernating boxes do not overlap — there is a clean ~280-day cutoff. This confirms the sharp recency split visible in the PCA scatter.

6.2 Frequency Distribution

- **Potential Loyalists and Hibernating:** Both have frequency = 1.0 for all members (the box collapses to a single line). They are one-time buyers.
- **Champions:** Frequency = 2.0 for all members (also a single line). They are the only repeat buyers in the dataset.
- **Key insight:** Frequency is binary in this dataset (1 or 2), which explains why Champions form a cleanly separated cluster in PCA space ($PC1 > 3$).

6.3 Monetary Distribution

- **Potential Loyalists:** Median ~100 BRL, IQR from ~50 to ~200. Moderate spenders.
- **Champions:** Widest distribution with median ~240 BRL, IQR from ~130 to ~370. Higher spending reflects their repeat-purchase behaviour.

- **Hibernating:** Nearly identical to Potential Loyalists (median ~100, similar IQR). This confirms that Potential Loyalists and Hibernating differ only on recency, not on spending behaviour.

Note: The monetary distributions of Potential Loyalists and Hibernating overlap almost completely. This is consistent with the SHAP analysis: monetary contributes < 2% of discriminative power because it does not help separate these two groups.

7. Customer Segments in PCA Space

The scatter plot below shows the final segmented customers projected into PCA space. This is the culmination of all four phases — customers now have both their PCA coordinates (for geometric understanding) and their segment labels (for business understanding).

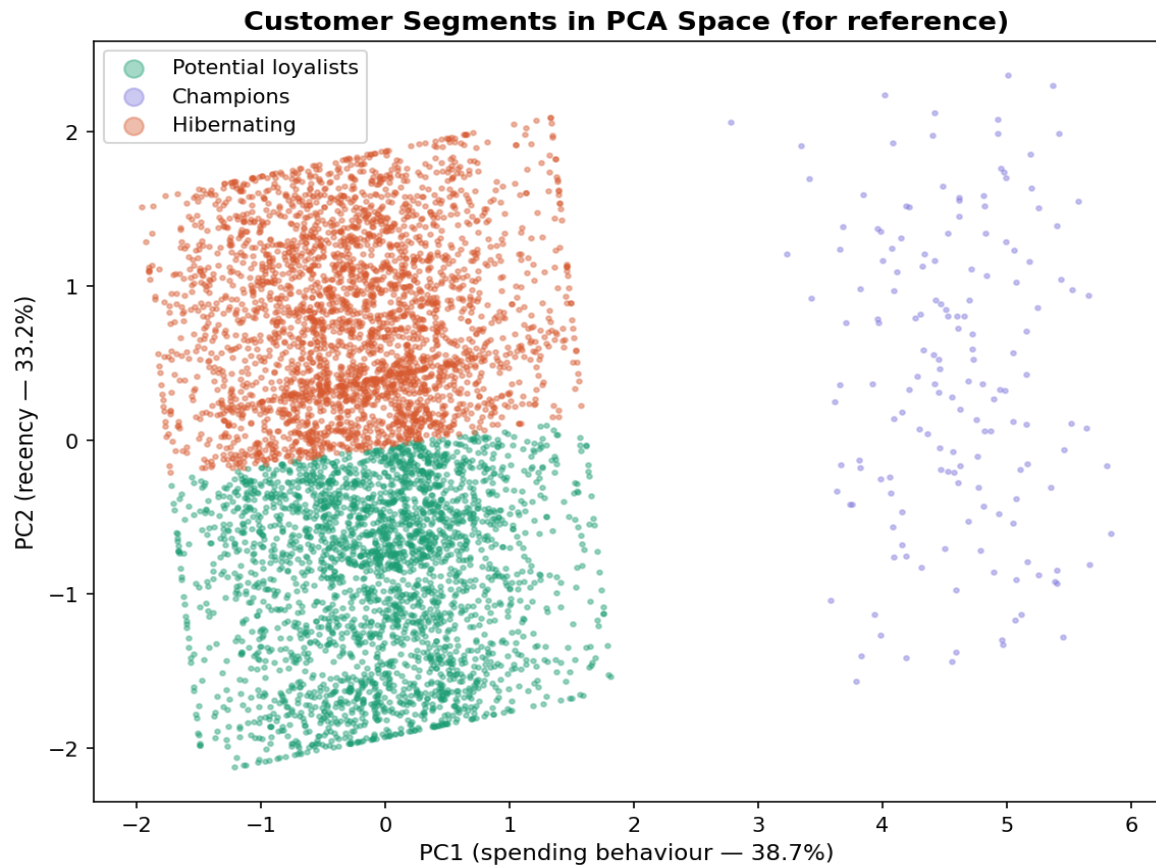


Figure 6: Customer segments in PCA space. Green = Potential Loyalists (bottom), Orange = Hibernating (top), Purple = Champions (right). The recency axis (PC2) separates the two large segments; the spending axis (PC1) separates Champions from the rest.

7.1 Segment Geometry

- **Potential Loyalists (green):** Bottom half of the main cluster body. $PC2 < 0$ corresponds to lower-than-average recency (more recent purchases). Spread across $PC1 \in [-2, 1.5]$ shows variation in spending.
- **Hibernating (orange):** Top half of the main body. $PC2 > 0$ corresponds to higher-than-average recency (older purchases). Same $PC1$ range as Potential Loyalists, confirming they differ only on the recency axis.
- **Champions (purple):** Isolated island at $PC1 > 3$. Separated from the main body along $PC1$ (spending behaviour), which is composed of $0.70 \times \text{frequency} + 0.70 \times \text{monetary}$ in the PCA loading vector.

7.2 Why the Segments Make Sense

The PCA scatter confirms the SHAP findings visually:

- **PC2 (recency — 33.2% variance)** is the primary axis separating the two large segments. SHAP assigns recency a dominance of 92–93% for these segments.
- **PC1 (spending — 38.7% variance)** separates Champions. SHAP assigns frequency a dominance of 90% for Champions, and frequency is the main component of PC1.
- The mathematical explanation (SHAP) and the geometric explanation (PCA scatter) are consistent — both point to recency as the primary differentiator.

8. Business Recommendations per Segment

The segment profiles and SHAP analysis translate directly into marketing strategy. Each segment has a distinct behavioural profile that suggests specific interventions.

8.1 Potential Loyalists (45.9% — 42,865 Customers)

Profile: Recent one-time buyers (155 days avg), moderate spend (BRL 154).

Opportunity: These customers are the easiest to convert into repeat buyers. They purchased recently and are still engaged.

- Send personalised follow-up emails with complementary product recommendations
- Offer a time-limited loyalty discount for their second purchase
- Implement a first-purchase-anniversary reminder campaign
- Track cohort conversion rate to Champions over 90/180/365 day windows

8.2 Champions (3.0% — 2,801 Customers)

Profile: Repeat buyers (frequency 2.0), highest spend (BRL 295), moderate recency (269 days).

Opportunity: The most valuable 3% — retention is the priority.

- Enrol in a VIP loyalty programme with exclusive early-access to deals
- Provide priority customer service and dedicated support channels
- Send personalised high-value product recommendations based on purchase history
- Monitor for churn signals: if recency exceeds 365 days, trigger a win-back campaign
- Calculate Customer Lifetime Value (CLV) for this segment to justify retention spend

8.3 Hibernating (51.1% — 47,691 Customers)

Profile: Lapsed one-time buyers (405 days avg), moderate spend (BRL 156).

Opportunity: The largest segment but hardest to re-engage. Cost-effectiveness is critical — not all Hibernating customers are worth pursuing.

- Segment further by recency: 280–365 days (warm) vs 365+ days (cold)
- Send a 'We miss you' win-back email with a compelling discount
- For cold customers (400+ days), consider a final re-engagement attempt before deprioritising
- A/B test different win-back offers (discount % vs free shipping vs new product alert)
- Track reactivation rate — if < 2%, the cost of outreach may exceed the revenue

8.4 Summary of Segment Strategy

Segment	Size	Priority	Strategy	KPI to Track
Potential Loyalists	45.9%	High	Convert to repeat buyers with targeted follow-ups	2nd purchase conversion rate
Champions	3.0%	Critical	Retain with VIP treatment and loyalty programmes	Retention rate, CLV
Hibernating	51.1%	Medium	Cost-effective win-back campaigns (sub-segment first)	Reactivation rate, CAC vs revenue

Note: These recommendations assume the Olist marketplace context (Brazilian e-commerce). Actual implementation should factor in: communication channel preferences (email, WhatsApp, SMS), regional buying patterns, product category affinity per segment, and the cost of each intervention vs expected uplift.